

Algorithms Problem Solving

Level 3

Split String

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the foreground, the silhouettes of several people are visible, some standing and some sitting, suggesting a collaborative workspace or a public area. The overall tone is professional and tech-oriented.

Algorithms Problem Solving

Level 3

Split String

Definition

Custom Delimiter String Splitting

-  Program divides a string into multiple tokens using a specified delimiter.

-  The extracted tokens are stored in a `vector<string>` for further processing.

Explanation

Delimiter Search

-  ``str.find(delim)`` searches for the next occurrence of the chosen delimiter.
-  The delimiter can contain one or multiple characters.

Token Extraction

-  ``substr()`` extracts the text located before the delimiter.
-  Each extracted token is pushed into the vector.

Position Advancement

-  After a token is extracted, the search position moves beyond the complete delimiter.
-  This is important when the delimiter contains multiple characters.

Final Token

-  After no more delimiters are found, the remaining text is added as the final token.
-  Without this step, the last word would be lost.

Procedures

- 1 Define ` SplitString(str, delim)` .
- 2 Search for the delimiter.
- 3 Extract the substring before it.
- 4 Store the token in the vector.
- 5 Move to the next search position.
- 6 Repeat until no delimiter remains.
- 7 Store the final remaining token.
- 8 Return the vector.

💡 Example

Input:

"Sara Alterkawi"

Delimiter:

" "

➡ Result

Tokens:

1: Sara

2: Alterkawi

Note

- ◆ The delimiter can be ``" "``, ``", "``, ``"##/"``, or another custom sequence.
- ◆ Empty tokens can be preserved or ignored depending on the implementation.
- ◆ The function is useful for parsing structured text records.

⊘ Warning

- ◆ Forgetting to push the final remaining substring loses the last token.
- ◆ ``delim.length()`` must be included when advancing the search position.
- ◆ Consecutive delimiters can produce empty tokens unless explicitly filtered.



Comparison

▨ Custom `SplitString()`

- ✓ Supports arbitrary delimiter strings.
- ✓ Stores the result in `vector<string>`.
- ✓ Works well with structured records such as `#//#`-separated files.

▤ `stringstream`

- ✓ Convenient for whitespace-based tokenization.
- ✓ Automatically skips standard whitespace.
- ✓ Is less direct for arbitrary multi-character delimiters.

✦ Summary

- ✓ Searches for configurable delimiters.
- ✓ Extracts each token using ``substr()``.
- ✓ Stores tokens inside a vector.
- ✓ Provides a reusable foundation for parsing text data.

Thank you 😊